DREXEL UNIVERSITY
Antoinette Westphal
College of Media Arts & Design
DEPARTMENT OF DIGITAL MEDIA

COURSE SYLLABUS

DIGM-553
Experimental Digital Media Lab

Spring 2026

3.0 credits

Instructor Information

Instructor: Darren Woodland, Jr.
Email: dkw34@drexel.edu
Office Hours: By appointment
Office Hour - Microsoft Bookings

Student Learning Information

Course Description

In this hands-on lab, students experiment with emerging tools and techniques, applying their creative vision to produce innovative digital media projects. This course encourages risk-taking and exploration, helping students develop unique projects that push the boundaries of digital art and design.

Prerequisites

None.

Course Purpose within a Program of Study

This course emphasizes experimentation with digital media tools, encouraging students to explore and push the boundaries of creative possibilities. As an essential part of the Art and Design domain, it allows students to apply emerging techniques in hands-on projects, cultivating a risk-taking mindset. This course provides the practical foundation that complements theoretical understanding, preparing students for more advanced experimentation and interdisciplinary collaboration in their future courses.

Statement of Expected Learning

- Develop experimental digital media projects through a research-creation methodology that integrates critical inquiry, technical exploration, and creative production as co-constitutive practices.
- Critically interrogate digital tools as non-neutral agents that encode worldviews and actively structure the boundaries of creative practice.
- Recognize and articulate the entanglement of practitioner, tool, and material in the creative process.
- Design and iterate enabling constraints that guide the creative process without predetermining outcomes.
- Articulate the theoretical and conceptual stakes of experimental media work through written, spoken, and performative modes of rendering research public.
- Collaborate with peers through techniques of relation that foreground process, care, and collective becoming.

Course Materials

Required and Recommended Texts, Readings, and Resources

Required Texts:

- Manning, E. & Massumi, B. (2014). *Thought in the Act: Passages in the Ecology of Experience*. University of Minnesota Press. [Propositions 0–3, pp. 83–94; ~12 pp]
- Magnusson, T. (2019). “Epistemic Tools.” In *Sonic Writing: Technologies of Material, Symbolic, and Signal Inscriptions*. Bloomsbury Academic. [Selected sections, ~8 pp]
- Agre, P. (1997). “Toward a Critical Technical Practice.” In *Computation and Human Experience*. Cambridge University Press. [pp. 151–155; ~5 pp]
- Loveless, N. (2019). “Haraway’s Dog.” In *How to Make Art at the End of the World: A Manifesto for Research-Creation*. Duke University Press. [pp. 19–29; ~10 pp]
- Camus, A. & Vinck, D. (2019). “Unfolding Digital Materiality.” In *Digitalsts*. Routledge. [Part 2, pp. 23–28; ~6 pp]

Recommended Readings:

- Salter, C. (2010). *Entangled: Technology and the Transformation of Performance*. MIT Press.

- Barad, K. (2003). "Posthumanist Performativity." *Signs*, 28(3).
- Guerra, J. & Ostergaard, S. (2017). "Technopoiesis as Complex Dynamic Knowledge Construction." *ICONO14*, 15(1).
- Sengers, P. et al. (2005). "Reflective Design." In *Proc. Critical Computing*. Aarhus, Denmark.

Readings to be provided by instructor

Required and Supplemental Materials and Technologies

Required Materials:

- Adafruit QT Py RP2040 (Product ID: 4900) with STEMMA QT cable (Product ID: 4210) [carried over from previous course].
- Adafruit 12-Key Capacitive Touch Sensor Breakout MPR121 (Product ID: 1982) [carried over from previous course].
- Premium Female/Male Extension Jumper Wires (Product ID: 1952) and USB-C data cable (Product ID: 5031).
- Computer with Drexel/Westphal's recommended specifications for running digital software.
- Access to at least one software environment for real-time interaction with sensor data (e.g., TouchDesigner, Max/MSP, Processing, p5.js, Pure Data, Unity).

Recommended Technologies:

- Adafruit ESP32 Feather V2 (Product ID: 5400) for wireless (WiFi/Bluetooth) projects.
- Additional STEMMA QT-compatible sensors and actuators from Adafruit (accelerometers, light sensors, NeoPixels, servos, etc.) as needed for individual projects.
- TouchDesigner for real-time interactive projects and prototyping.
- Platforms like Unity, Unreal Engine, Blender, or SuperCollider for experimenting with emerging technologies.
- Soldering iron and solder wire for hardware modifications.

Component availability varies by vendor. Students unable to source required hardware should contact the instructor to discuss alternatives prior to the relevant course week.

Assignments and Assessments

Graded Assignments and Learning Activities

Project Foundations: Potluck + Speculative Proposal (10%)
This bundled assessment comprises the early foundational exercises of the course. It includes your contribution to the Week 2 *(Im)material Potluck* and your formal **Speculative Proposal** (living document). The Proposal articulates: (a) your driving curiosity (your "dog"); (b) selected enabling constraints; (c) the matter-network of tools/materials; and (d) your Epistemic Positioning Statement. A Revised Proposal reflecting project transformation is due week 10.

- **Includes:** Week 2 Potluck + Week 4 Proposal + Revised Proposal (end of term).
- **Evaluation Criteria:** Conceptual rigor, specificity of constraints, tool analysis.
- **Submission:** PDF uploaded to Blackboard.

Critical Technical Journal (25%)
Throughout the course, maintain a journal documenting your making process and reflexive critique. This journal captures all smaller in-class deliverables such as the *Epistemic Tool Audit* (Week 3) and the *First Impasse Report* (Week 5). Dimensions of the journal: (1) Technical Documentation; (2) Epistemic Interrogation; (3) Entanglement Mapping.

- **Includes:** Installment 1 (Weeks 1–4, includes Epistemic Audit), Installment 2 (Weeks 5–7, includes Impasse Report), Installment 3 (Weeks 8–10, includes Process Seed Reflection).
- **Evaluation Criteria:** Depth of engagement, documentation quality, honest tracking of failure/pivot.
- **Submission:** Digital format (PDF) at three specific check-points. However, actual journal can take many forms including physical notebook, blog, digital whiteboard, etc. *Format can be negotiated with professor*

Experimental Media Project (35%)
Develop an original digital media project that demonstrates "thinking-in-action." Projects are encouraged to utilize physical computing hardware entangled with software environments. Evaluated on the rigor of the path taken and not just final polish.

- **Includes:** Initial prototyping, Mid-term status check (see below), and Final Project Submission.
- **Evaluation Criteria:** Creative risk, conceptual-material integration, technical engagement.
- **Submission:** Final build/documentation uploaded Week 10.

Structured Dialogue + Participation (10%)
Participation in two formal review sessions (Weeks 6 & 9). This grade also encompasses your presentation at the **Mid-term Review** (Week 7) and larger class engagement. Critique sessions utilize specific formats (e.g. Six Thinking Hats) to focus on activating potential rather than judgment.

- **Includes:** Participation in critiques + Mid-term presentation delivery.
- **Evaluation Criteria:** Generosity, quality of engagement, application of critical lenses.

Final Presentation (20%)
Render your work public. This presentation (Week 11) should expose the formative forces of the project. This should not be just: "what you made," but must also include how the encounter with tools/materials changed the inquiry.

- **Submission:** 12–15 min presentation delivering narrative of process + output.

Submission Information

All assignments should be submitted through Blackboard Learn Assignments unless otherwise specified. File formats should be PDF, video (MP4), or other specified formats. Please refer to each assignment's specific guidelines for additional details.

Grading Matrix

Assignment	Criteria	Weight (%)	Exemplary (90-100%)	Proficient (80-89%)	Developing (70-79%)	Insufficient (Below 70%)
Proposal + Epistemic Positioning	Sharpness, constraints, tool analysis	10%	Exceptional framing and constraints. Clear processual transformation.	Accurate constraints and positioning; lacks deep specificity.	Generic constraints. Shallow epistemic awareness.	Lacks rigor, constraints, or repositioning.
Critical Technical Journal	Reflexive depth across 3 areas	25%	Rigorous focus on tech, epistemology, and entanglement. Detailed and rich.	Consistent documentation but dimensions are underdeveloped.	Inconsistent entries. Surface-level reflexive questioning.	Incomplete or purely descriptive log.
Experimental Media Project	Conceptual-Material integration	35%	Total integration of inquiry and making. High creative/technical risk.	Strong execution; conceptual integration could be deeper.	Basic execution. Small creative risk or shallow inquiry.	technically negligible or lacks core RC method.
Six Hats dialogue + Reflection	Engagement, Lenses, Care	10%	Active and nuanced use of all hats/lenses. Collective thinking focused.	Good participation with safe, less-engaged feedback.	Minimal participation. Purely reactive or shallow feedback.	Absent or disengaged.
Final Presentation	Rendering the work public	20%	Clear rendering of process and formative forces. Formally inventive.	Adequate process narrative but lacks formal invention.	Describes results only; process/sticks are invisible.	Unclear, unprepared, or misses the conceptual stakes.

Grade Scale

Grade	Percentage	Description
A+	100%+	Outstanding Work
A	95-99%	Very Superior Work
A-	90-94%	Superior Work
B+	88-89%	Above Good Work
B	83-87%	Good Work
B-	80-82%	Slightly Below Good Work
C+	78-79%	Slightly Above Average Work
C	73-77%	Average Work
C-	70-72%	Slightly Below Average Work
D+	68-69%	Below Average Work
D	63-67%	Below Average Work
F	00-62%	Failure. Work Unacceptable

Course Calendar

Week	Learning Activities	Assessment
1	**Practice Immanent Critique.** Course introduction: research-creation, epistemic tools, and entanglement. Introduce the (Im)material Potluck and Critical Technical Journal. Hardware/software/material inventory.	Read Manning & Massumi, Propositions 0–3 for next week.
2	**Construct Conditions for Speculative Pragmatism.** Discussion of enabling constraints. (Im)material Potluck presentations. Enabling Constraints Workshop using potluck materials and Adafruit hardware.	Journal Entry 0: Potluck reflection. Read Magnusson + Agre for next week.
3	**Epistemic Tools + Entanglement.** Concept drop: Magnusson's epistemic tools, Agre's split identity, Barad's entanglement. The Epistemic Tool Audit workshop (software + hardware). Hands-on exploration at the edges of your tools.	Journal Entry 1: Epistemic Tool Audit. Read Loveless pp. 19–29 for next week.
4	**Design Enabling Constraints.** Discussion: "What is your dog?" Peer workshopping of proposals.	**Speculative Proposal due. Journal Installment 1 due.** Read Camus & Vinck Part 2 for next week.
5	**Enact Thought.** Brief discussion of Camus & Vinck on becoming digital and matter-networks. Studio time: deep making begins. Identify first impasse.	Journal Entry 2: First impasse report.
6	**Invent Platforms for Relation.** Collective Critique Session 1. Studio time responding to critique.	Critique reflection (half page).
7	**Embrace Failure.** Mid-term presentations: what isn't working and what it reveals.	**Mid-term presentation. Journal Installment 2 due.**
8	**Practice Letting Go.** Process Seed Exchange. Studio time: revision as transformation.	Journal Entry 3: Process seed reflection.
9	**Render Formative Forces.** Collective Critique Session 2. Studio time: final development.	Critique reflection (half page).
10	**Creatively Return to Chaos.** Final studio time. One-on-one check-ins. Finalize all materials.	**Journal Installment 3 due.**
11	**Proceed.** Final presentations: rendering the work and its process public. Collective reflection.	**Final Project due. Revised Speculative Proposal due. Final Presentation. TURNED IN ONLINE**

Academic Policies

Absence from Class Policy: For information regarding class attendance and absences, please refer to the Drexel University Absence from Class Policy. Students are expected to communicate with their instructors about any absences and adhere to the guidelines outlined.

Academic Integrity, Plagiarism, Dishonesty, and Cheating Policy: Detailed information about academic integrity, including policies on plagiarism and cheating, can be found in the Drexel University Academic Integrity Policy. Upholding honesty in all academic work is imperative, and violations will be addressed according to university procedures.

Permitted Use of Artificial Intelligence Tools in This Course: The use of AI tools, such as ChatGPT, is permitted for enhancing creative exploration and speculative design within this course. Students must use AI tools transparently, cite them appropriately, and ensure their contributions are genuine. Refer to the Academic Integrity Pertaining to Artificial Intelligence Policy for more details.

Course Add/Drop Policy: Information about adding or dropping courses is available in the Course Add/Drop Policy. Adherence to university-specified deadlines is crucial.

Course Withdrawal Policy: For guidelines on withdrawing from courses, consult the Course Withdrawal Policy. Ensure compliance with the procedures and deadlines specified.

Reporting Title IX (Sexual Harassment and Misconduct), Discrimination, Harassment, and Bias Incidents: To report incidents related to sexual harassment, misconduct, discrimination, harassment, or bias, refer to the Title IX Policy. The university is committed to maintaining a safe and inclusive environment.

Student with Disability Statement: Students seeking accommodations due to disabilities should contact the Office of Disability Resources promptly. This office provides support to ensure equal access to all learning opportunities.

Student Conduct and Care: All students are expected to adhere to Drexel's standards of student conduct. Details can be found in the Student Handbook. Violations of the code of conduct may result in disciplinary action.

Class Recording: Meetings of this course might be recorded. Any recordings will be available to students registered for this class. Students are expected to follow appropriate university policies and maintain the security of passwords used to access recorded lectures. Recordings, or any part thereof, may not be reproduced, shared with those not in the class, or uploaded to other online environments.

Appropriate Use of Course Materials Policy: Course materials provided may be the intellectual property of Drexel University, the course instructor, or others. Use of this intellectual property is governed by Drexel University's policies, including the Appropriate Use of Course Materials Policy. Unauthorized reproduction, distribution, or reposting of course materials is prohibited and may be considered a violation of university policies.

Instructor Content Adjustment Clause

The instructor delivering this course reserves the right to tailor the course content, including readings, assignments, and assessment methods, to best suit the needs of the students and the evolving nature of the field. While the course description, title, and course purpose will remain as stated, instructors may modify specific learning activities, materials, and assignment requirements to enhance the learning experience. These modifications will always align with the course's overall objectives and learning outcomes as outlined in this syllabus.